

Plant Disease Detection Using Convolutional Neural Networks

Presented by -

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Abstract

One of the most important pillars of human survival is agriculture, and plant diseases directly affect the productivity and security of food. Large-scale agricultural damage and financial loss can be avoided by the use of early identification and precise plant disease classification. Convolutional Neural Networks (CNN) will be used in this research to create a deep learning model that will automatically identify and categorize diseases of plants from photos of leaves. A dataset of both healthy and damaged plant leaves is used to train the suggested CNN model. The model's encouraging validation accuracy showed CNNs' potential for automated illness diagnosis. Transfer learning using ResNet50 will be used in subsequent studies to improve classification robustness and accuracy even more.

Introduction

Globally, plant diseases have a major impact on food supply, quality, and agricultural productivity. Conventional disease detection techniques necessitate laboratory analysis and specialized knowledge, which are costly, time-consuming, and frequently impractical for extensive surveillance. Convolutional Neural Networks (CNNs), a subset of Deep Learning (DL), have transformed picture classification jobs in recent years by allowing computers to directly learn intricate visual patterns from unprocessed photos.

CNNs are used in this study to recognize and categorize illnesses from photos of plant leaves. Manual feature extraction is no longer necessary because the CNN model automatically extracts discriminative characteristics from the input images. To help farmers and agricultural researchers detect diseases quickly, the main goal is to create a model that can reliably distinguish between healthy and diseased plants.

Problem Statement

Plant disease detection through manual observation is highly inefficient for large-scale farming and often subject to human error. Hence, there is a need for an automated, accurate, and scalable system that can classify plant diseases from images.

Objectives

- To pre-process a dataset of plant leaf images for model training and validation.
- To build and train a custom CNN architecture for disease classification.
- To evaluate model performance using accuracy and loss metrics.

Methodology

The proposed system for Plant Disease Detection using Convolutional Neural Networks (CNN) follows a structured pipeline consisting of data pre-processing, model architecture design, training and validation, and performance evaluation.

1. Dataset Preparation

The dataset used in this project consists of images of plant leaves belonging to different disease categories as well as healthy plants.

Each image is labelled according to the type of disease or health condition. The dataset is divided into training and validation subsets to enable unbiased evaluation of model performance.

Steps Involved:

I. Data Loading:

The dataset was loaded using PyTorch's DataLoader and ImageFolder utilities for automatic image labeling based on directory structure.

II. Image Preprocessing:

Each image was resized to a fixed dimension to maintain consistency across the dataset.

III. Normalization:

Pixel values were normalized to a [0,1] or [-1,1] range to stabilize and speed up convergence during training.

IV. Data Augmentation:

To increase the diversity of the training data and improve generalization, random transformations such as:

- Horizontal/Vertical flipping
 - Rotation
 - Zooming
 - Shifting
- were applied using "torchvision.transforms".

2. CNN Model Architecture

A **custom CNN architecture** was implemented from scratch using the PyTorch deep learning framework. The architecture was designed to automatically extract spatial and texture-based features from the leaf images.

LAYER (TYPE)	OUTPUT SHAPE	PARAM #
CONV2D-1	[-1, 32, 128, 128]	896
MAXPOOL2D-2	[-1, 32, 64, 64]	0
CONV2D-3	[-1, 64, 64, 64]	18,496
MAXPOOL2D-4	[-1, 64, 32, 32]	0
LINEAR-5	[-1, 256]	16,777,472
DROPOUT-6	[-1, 256]	0
LINEAR-7	[-1, 38]	9,766

3. Training Setup

The model was trained using a supervised learning approach.

Parameters:

a. Loss Function:

nn.CrossEntropyLoss() was used, which is equivalent to Categorical Cross-Entropy and suitable for multi-class classification.

b. Optimizer:

The Adam optimizer with LR 0.001 was chosen for its adaptive learning rate capabilities, providing fast and stable convergence.

c. Learning Rate Scheduler:

A ReduceLROnPlateau scheduler was implemented to reduce the learning rate when validation performance plateaued.

4. Training Procedure:

- I. The model was trained for **50 epochs** with a **batch size** optimized based on available GPU memory.
- II. For each epoch:
 - The training loop computed **loss** and **accuracy** over batches.
 - Validation was performed after each epoch to monitor generalization performance.
- III. **Early Stopping** and **Model Checkpointing** were applied:
 - Early stopping halted training when validation accuracy stopped improving for 7 consecutive epochs.
 - The best-performing model weights were saved as best_model.pth.

5. Performance Evaluation

To monitor model performance, the following metrics were calculated after every epoch:

- Training Loss and Validation Loss – to observe convergence and overfitting trends.
- Training Accuracy and Validation Accuracy – to assess predictive capability on unseen data.

Graphs of loss vs. epoch and accuracy vs. epoch were plotted to visualize learning behaviour. The model achieved stable training and validation curves, indicating effective learning.

Results and Analysis

The CNN model successfully learned the visual features of diseased and healthy leaves.

- The **training accuracy** of 89.76% showed a consistent increase over epochs.
- The **validation accuracy** also improved and stabilized before early stopping occurred. Final Validation accuracy achieved was 95.62%.
- The model achieved a **high validation accuracy**, demonstrating effective generalization on unseen data.

Although the model performed well, there was a slight gap between training and validation accuracy, indicating potential overfitting or data imbalance, which will be addressed in future improvements.

Future Work

To further enhance the accuracy and efficiency of the model, transfer learning using ResNet50 (a pre-trained convolutional neural network trained on ImageNet) will be implemented.

This approach is expected to:

- Improve feature extraction through pre-learned representations.
- Reduce training time.
- Achieve better accuracy and robustness with limited data.

Conclusion

In this study, a Convolutional Neural Network (CNN) model was used to successfully build a deep learning-based method for plant disease identification. The system showed that it could automatically recognize discriminative visual characteristics in photos of plant leaves and correctly categorize them as either healthy or unhealthy. The model successfully captured the spatial and textural features necessary for disease recognition, according to the training and validation performance measures.